Difference Equation Descriptions System Characteristics

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Key Concepts

1) The solution to a difference equation gives insight into the relationship between the difference equation parameters and the corresponding system characteristics.

2) Difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]; \qquad a_0 = 1$$

3) Output $y[n] = y_s[n] + y_t[n]$

a) Steady-state response $y_s[n]$. If $x[n] = z^n$ then $y_s[n] = H(z)z^n$ where

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

b) Transient response $y_t[n] = \sum_{\ell=1}^{N} c_\ell d_\ell^n$ where d_ℓ are the N (distinct) roots of the characteristic equation

$$\sum_{k=0}^{N} a_k z^{-k} = 0$$

and c_ℓ are chosen so $y[n]$ satisfies initial conditions $y[0], y[1], \ldots, y[N-1]$.

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4) Stable systems have $|d_\ell| < 1$ for all ℓ. If $|d_\ell| > 1$, then $|y[n]|$ can grow without bound due to $d_\ell^n \to \infty$ as $n \to \infty$.

5) Under steady-state conditions the output of the system is the same form as the input. For example, sinusoidal input produces a sinusoidal output of the same frequency.

6) The transient response describes how the system transitions from its initial state (at rest or having stored energy) to steady-state behavior. It depends on d_ℓ, the roots of the characteristic equation.

    a) If d_ℓ is real, then it contributes a decaying exponential to the output.

i) If d_ℓ is negative, the sign of the decaying exponential alternates.

ii) The response decays slowly for $|d_\ell| \approx 1$ and decays quickly for $|d_\ell| \approx 0$

b) If d_ℓ is complex, it contributes exponentially damped sinusoidal terms to the output.

i) The rate of exponential decay depends on the magnitude of d_ℓ with slow decay for $|d_\ell| \approx 1$ and fast decay for $|d_\ell| \approx 0$.

ii) The phase of d_ℓ determines the frequency of the exponentially damped sinusoid.

iii) If the coefficients of the difference equation are real valued, then the d_ℓ occur in complex conjugate pairs.

c) The magnitude of the roots of the characteristic equation ($|d_\ell|$) determine the response time of the system. Response time decreases as $|d_\ell|$ decreases. Systems with a $|d_\ell| \approx 1$ generally have slow response times.

Video Screenshots

The method for finding the output of a system described by a difference equation provides insight

Summary

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \; ; \; a_0 = 1$$

Output has two parts $y[n] = y_s[n] + y_t[n]$

$y_s[n]$: steady state; $y_t[n]$: transient response

1) Steady-state response

if $x[n] = z^n$, then $y_s[n] = H(z) z^n$ for z a complex number

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

2) Transient response

$$y_t[n] = \sum_{l=1}^{N} C_l (d_l)^n$$

where d_l are the N roots of $\sum_{k=0}^{N} a_k z^{-k} = 0$, C_l chosen so $y[n]$ meets initial conditions

Insights into System Behavior

Steady-state conditions

- stored energy dissipates
- effect of system being at rest is overcome

Output takes the same form as the input!

$$x[n] = z^n \xrightarrow[n\to\infty]{} y[n] = H(z)\,z^n$$

transient response → 0 for stable systems.

Examples –

1) Complex sinusoid $z = e^{j\omega} \longrightarrow y[n] = H(e^{j\omega})\,e^{j\omega n}$

2) Real sinusoid $x[n] = \cos(\omega n) = \frac{1}{2}(e^{j\omega})^n + \frac{1}{2}(e^{-j\omega})^n$

$$y[n] = \frac{1}{2}H(e^{j\omega})e^{j\omega n} + \frac{1}{2}H(e^{-j\omega})(e^{-j\omega})^n = |H(e^{j\omega})|\cos(\omega n + \arg\{H(e^{j\omega})\})$$

Transient Response

How does the system transition from its initial state (stored energy or at rest) to steady-state behavior?

$$y_t[n] = \sum_{\ell=1}^{N} c_\ell (d_\ell)^n$$

c_ℓ depend on initial conditions

The transient behavior depends on roots d_ℓ of

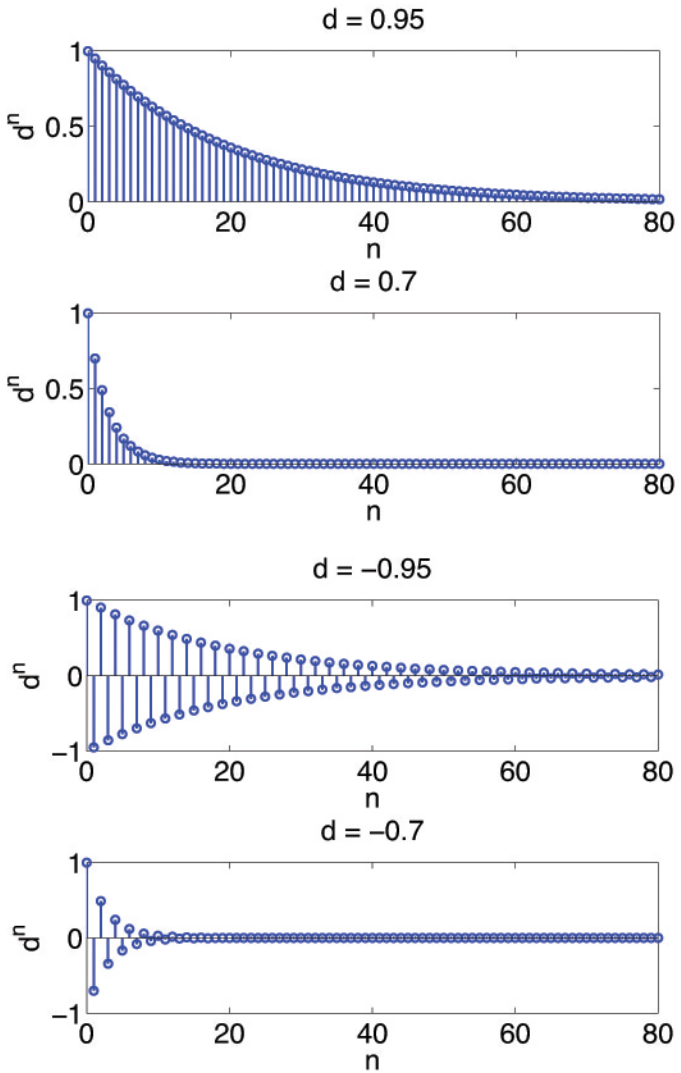
$$\sum_{k=0}^{N} a_k z^{-k} = 0 \qquad \left(\text{or} \quad \sum_{k=0}^{N} a_k z^{N-k} = 0\right)$$

What does d^n look like?

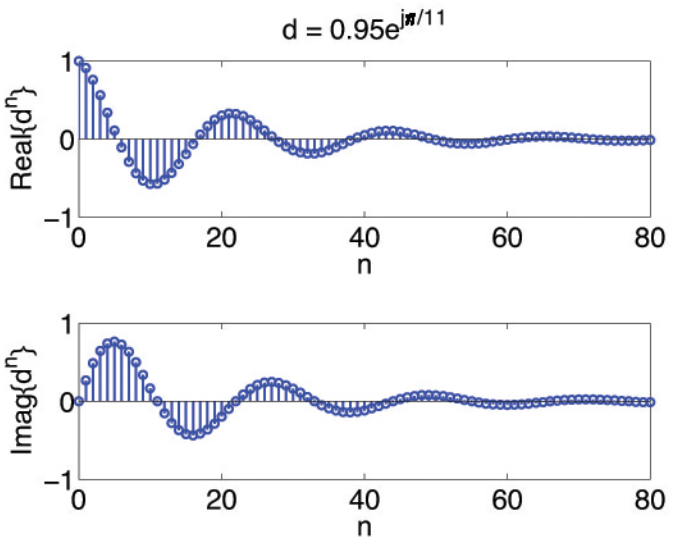
Depends on $|d|$ and whether d is real or complex

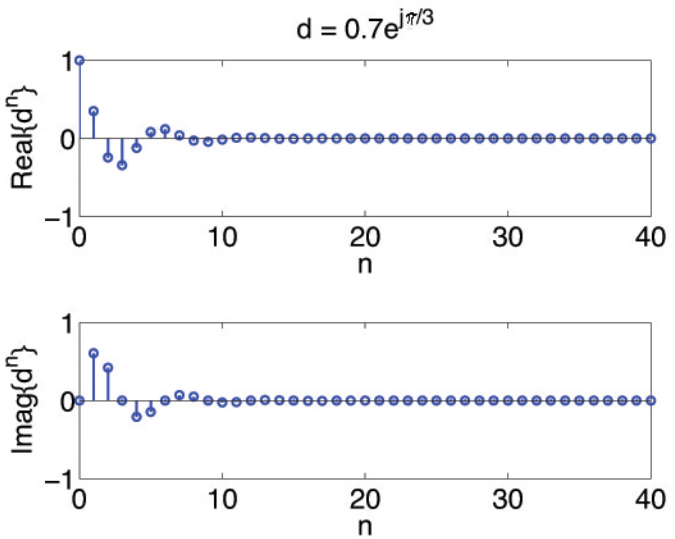
Examples of d^n ($|d|<1$)

d real



d complex





Difference Equation Solutions

$x[n] = u[n]$ Initial Conditions = 0

$y[n] - 0.95\, y[n-1] = x[n]$

$y[n] + a_1 y[n-1] + a_2 y[n-2] = x[n]$

